API Reference

Get Quote

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This API returns an executable quote for swapping, bridging and calling

POST https://api.relay.link / quote / v2

Try it

cURL

cURL

```
curl --request POST \
  --url https://api.relay.link/quote/v2 \
  --header 'Content-Type: application/json' \
  --data '
{
  "user": "0x03508bb71268bba25ecacc8f620e01866650532c",
  "originChainId": 8453,
  "destinationChainId": 10,
  "originCurrency": "0x0000000000000000000000000000000000000000",
  "destinationCurrency": "0x0000000000000000000000000000000000000000",
  "amount": "1000000000000000000",
  "tradeType": "EXACT_INPUT"
}
'
```

Body

application/json

200 400 401 429 500

user string required

Address that is depositing funds on the origin chain and submitting transactions or signatures

originChainId number required

destinationChainId number required

originCurrency string required

destinationCurrency string required

amount string required

Amount to swap as the base amount (can be switched to exact input/output using the dedicated flag), denoted in the smallest unit of the specified currency (e.g. wei for ETH)

Pattern: `^[0-9]+$`

tradeType enum<string> required

Whether to use the amount as the output or the input for the basis of the swap

Learn more about trade types

Available options: `EXACT_INPUT`, `EXACT_OUTPUT`, `EXPECTED_OUTPUT`

recipient string

Address that is receiving the funds on the destination chain, if not specified then this will default to the user address

txs object[]


```
{
  "steps": [
    {
      "id": "deposit",
      "action": "Confirm transaction in your wallet",
      "description": "Depositing funds to the relayer to execute the swap for USDC",
      "kind": "transaction",
      "requestId": "0x92b99e6e1ee1deeb9531b5ad7f87091b3d71254b3176de9e8b5f6c6d0bd3a331",
      "items": [
        {
          "status": "incomplete",
          "data": {
            "from": "0x0CccD55A5Ac261Ea29136831eeaA93bfE07f5Db6",
            "to": "0xf70da97812cb96acdf810712aa562db8dfa3dbef",
            "data": "0x00fad611",
            "value": "1000000000000000000",
            "maxFeePerGas": "12205661344",
            "maxPriorityFeePerGas": "2037863396",
            "chainId": 1
          },
          "check": {
            "endpoint": "/intents/status?requestId=0x92b99e6e1ee1deeb9531b5ad7f87091b3d71254b
            "method": "GET"
          }
        }
      ]
    }
  ],
  "fees": {
    "gas": {
      "currency": {
        "chainId": 8453,
        "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
        "symbol": "USDC",
        "name": "USD Coin",
        "decimals": 6,
        "metadata": {
          "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
          "verified": false,
          "isNative": false
        }
      }
      "amount": "30754920",
```

```
        "amountFormatted": "30.75492",
        "amountUsd": "30.901612",
        "minimumAmount": "30454920"
      },
      "relayer": {
        "currency": {
          "chainId": 8453,
          "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
          "symbol": "USDC",
          "name": "USD Coin",
          "decimals": 6,
          "metadata": {
            "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
            "verified": false,
            "isNative": false
          }
        },
        "amount": "30754920",
        "amountFormatted": "30.75492",
        "amountUsd": "30.901612",
        "minimumAmount": "30454920"
      },
      "relayerGas": {
        "currency": {
          "chainId": 8453,
          "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
          "symbol": "USDC",
          "name": "USD Coin",
          "decimals": 6,
          "metadata": {
            "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
            "verified": false,
            "isNative": false
          }
        },
        "amount": "30754920",
        "amountFormatted": "30.75492",
        "amountUsd": "30.901612",
        "minimumAmount": "30454920"
      },
      "relayerService": {
        "currency": {
          "chainId": 8453,
          "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
          "symbol": "USDC",
```


Show child attributes

txsGasLimit number

Total gas limit for the destination chain call transactions

authorizationList object[]

Authorization list for EIP-7702 transactions to be executed on destination chain

Show child attributes

additionalData object

Additional data needed for specific routes

Show child attributes

referrer string

referrerAddress string

Pattern: `^0x[a-fA-F0-9]{40}$`

refundTo string

Address to send the refund to in the case of failure, if not specified then the recipient address or user address is used

refundOnOrigin boolean deprecated

Always refund on the origin chain in case of any issues

topupGas boolean

If set, the destination fill will include a gas topup to the recipient (only supported for EVM chains if the requested currency is the gas currency on the destination chain)

Learn more about topping up gas

topupGasAmount string

The destination gas topup amount in USD decimal format, e.g 100000 = $1. topupGas is required to be enabled. Defaults to 2000000 ($2)

useReceiver boolean default:true

Enable this to route payments via a receiver contract. This contract will emit an event when receiving payments before forwarding to the solver. This is needed when depositing from a smart contract as the payment will be an internal transaction and detecting such a transaction requires obtaining the transaction traces.

enableTrueExactOutput boolean default:false

Enabling will send any swap surplus when doing exact output operations to the solver EOA, otherwise it will be swept to the recipient

explicitDeposit boolean default:true

Enable this to avoid direct transfers to the depository (only relevant for EVM and v2 protocol flow)

useExternalLiquidity boolean

Enable this to use canonical bridging, trading speed for more liquidity

useFallbacks boolean

Enable this for specific fallback routes


```
"name": "USD Coin",
"decimals": 6,
"metadata": {
  "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
  "verified": false,
  "isNative": false
}
},
"amount": "30754920",
"amountFormatted": "30.75492",
"amountUsd": "30.901612",
"minimumAmount": "30454920"
},
"app": {
"currency": {
  "chainId": 8453,
  "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
  "symbol": "USDC",
  "name": "USD Coin",
  "decimals": 6,
  "metadata": {
    "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
    "verified": false,
    "isNative": false
  }
},
"amount": "30754920",
"amountFormatted": "30.75492",
"amountUsd": "30.901612",
"minimumAmount": "30454920"
},
"subsidized": {
"currency": {
  "chainId": 8453,
  "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
  "symbol": "USDC",
  "name": "USD Coin",
  "decimals": 6,
  "metadata": {
    "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
    "verified": false,
    "isNative": false
  }
},
"amount": "30754920",
```

usePermit boolean

Enable this to use permit (eip-2612) when bridging, only works on supported currency such as usdc

permitExpiry number

How long the permit remains valid, in seconds. Defaults to 10 minutes.

useDepositAddress boolean

Enable this to use a deposit address when bridging, in scenarios where calldata cannot be sent alongside the transaction. only works on native currency bridges.

strict boolean

When used with useDepositAddress on BVM or SVM chains, enables non-custodial deposit addresses. The solver never holds the private key.

slippageTolerance string

Slippage tolerance for the swap, if not specified then the slippage tolerance is automatically calculated to avoid front-running. This value is in basis points (1/100th of a percent), e.g. 50 for 0.5% slippage

latePaymentSlippageTolerance string

Slippage tolerance for destination gas in the event that the deposit occurs after the order deadline, and more gas is required for the solver to execute the destination transaction.

appFees object[]

Show child attributes

Learn more about app fees

gasLimitForDepositSpecifiedTxs number


```
      "amount": "30754920",
      "amountFormatted": "30.75492",
      "amountUsd": "30.901612",
      "minimumAmount": "30454920"
    }
  },
  "details": {
    "operation": "<string>",
    "sender": "<string>",
    "recipient": "<string>",
    "currencyIn": {
      "currency": {
        "chainId": 8453,
        "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
        "symbol": "USDC",
        "name": "USD Coin",
        "decimals": 6,
        "metadata": {
          "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
          "verified": false,
          "isNative": false
        }
      },
      "amount": "30754920",
      "amountFormatted": "30.75492",
      "amountUsd": "30.901612",
      "minimumAmount": "30454920"
    },
    "currencyOut": {
      "currency": {
        "chainId": 8453,
        "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
        "symbol": "USDC",
        "name": "USD Coin",
        "decimals": 6,
        "metadata": {
          "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
          "verified": false,
          "isNative": false
        }
      },
      "amount": "30754920",
      "amountFormatted": "30.75492",
      "amountUsd": "30.901612",
      "minimumAmount": "30454920"
    },
```

If the request involves specifying transactions to be executed during the deposit transaction, an explicit gas limit must be set when requesting the quote

forceSolverExecution `boolean`

Force executing swap requests via the solver (by default, same-chain swap requests are self-executed)

subsidizeFees `boolean`

If the sponsor should pay for the fees associated with the request. Includes gas topup amounts.

maxSubsidizationAmount `string`

The max subsidization amount in USDC decimal format, e.g 1000000 = $1. subsidizeFees must be enabled. This amount is the threshhold where if its surpassed the entire request will not be subsidized at all.

subsidizeRent `boolean`

If the sponsor should pay for the solana rent associated with the request.

includedSwapSources `string[]`

Swap sources to include for swap routing.

excludedSwapSources `string[]`

Swap sources to exclude for swap routing.

includedOriginSwapSources `string[]`

Swap sources to include for swap routing on origin.

includedDestinationSwapSources `string[]`

Swap sources to include for swap routing on destination.


```
"refundCurrency": {
  "currency": {
    "chainId": 8453,
    "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
    "symbol": "USDC",
    "name": "USD Coin",
    "decimals": 6,
    "metadata": {
      "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
      "verified": false,
      "isNative": false
    }
  },
  "amount": "30754920",
  "amountFormatted": "30.75492",
  "amountUsd": "30.901412",
  "minimumAmount": "30454920"
},
"timeEstimate": {
  "currency": {
    "chainId": 8453,
    "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
    "symbol": "USDC",
    "name": "USD Coin",
    "decimals": 6,
    "metadata": {
      "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
      "verified": false,
      "isNative": false
    }
  },
  "amount": "30754920",
  "amountFormatted": "30.75492",
  "amountUsd": "30.901612",
  "minimumAmount": "30454920"
},
"totalImpact": {
  "usd": "<string>",
  "percent": "<string>"
},
"swapImpact": {
  "usd": "<string>",
  "percent": "<string>"
},
"expandedPriceImpact": {
```

originGasOverhead `number`

The gas overhead for the origin chain, this is used to calculate the gas fee for the origin chain when the solver is executing a gasless transaction on the origin chain

depositFeePayer `string`

The payer to be set for deposit transactions on solana. This account must have enough for fees and rent.

maxRouteLength `number`

Maximum number of hops to use in solana swap routing. Can reduce transaction size.

useSharedAccounts `boolean`

Prevents certain ATA creation instructions in solana routing

includeComputeUnitLimit `boolean`

Whether to include compute unit limit instruction for solana origin requests.

overridePriceImpact `boolean`

Whether to ignore price impact errors.

disableOriginSwaps `boolean`

Whether to disable origin swaps.

fixedRate `string`

The rate to charge for fixed spread quotes.


```
        "swap": {
          "usd": "<string>"
        },
        "execution": {
          "usd": "<string>"
        },
        "relay": {
          "usd": "<string>"
        },
        "app": {
          "usd": "<string>"
        },
        "sponsored": {
          "usd": "<string>"
        }
      },
      "rate": "<string>",
      "slippageTolerance": {
        "origin": {
          "usd": "<string>",
          "value": "<string>",
          "percent": "<string>"
        },
        "destination": {
          "usd": "<string>",
          "value": "<string>",
          "percent": "<string>"
        }
      },
      "timeEstimate": 123,
      "userBalance": "<string>",
      "fallbackType": "<string>",
      "isFixedRate": true,
      "fixedRateFee": {
        "usd": "<string>"
      },
      "route": {
        "origin": {
          "inputCurrency": {
            "currency": {
              "chainId": 8453,
              "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
              "symbol": "USDC",
              "name": "USD Coin",
              "decimals": 6,
```

Response

200 application/json

Default Response

steps `object[]`

An array of steps detailing what needs to be done to bridge, steps includes multiple items of the same kind (signature, transaction, etc)

Show child attributes

Example:


```
[
  {
    "id": "deposit",
    "action": "Confirm transaction in your wallet",
    "description": "Depositing funds to the relayer to execute the swap for USDC",
    "kind": "transaction",
    "requestId": "0x92b99e6e1ee1deeb9531b5ad7f87091b3d71254b3176de9e8b5f6c6d0bd3a331",
    "items": [
      {
        "status": "incomplete",
        "data": {
          "from": "0x0CccD55A5Ac261Ea29136831eeaA93bfE07f5Db6",
          "to": "0xf70da97812cb96acdf810712aa562db8dfa3dbef",
          "data": "0x00fad611",
          "value": "1000000000000000000",
          "maxFeePerGas": "12205661344",
          "maxPriorityFeePerGas": "2037863396",
          "chainId": 1
        },
        "check": {
          "endpoint": "/intents/status?requestId=0x92b99e6e1ee1deeb9531b5ad7f87091b3d712
          "method": "GET"
        }
      }
    ]
  }
]
```



```
"metadata": {
  "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
  "verified": false,
  "isNative": false
}
},
"amount": "30754920",
"amountFormatted": "30.75492",
"amountUsd": "30.901612",
"minimumAmount": "30454920"
},
"outputCurrency": {
"currency": {
"chainId": 8453,
"address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
"symbol": "USDC",
"name": "USD Coin",
"decimals": 6,
"metadata": {
"logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
"verified": false,
"isNative": false
}
"amount": "30754920",
"amountFormatted": "30.75492",
"amountUsd": "30.901612",
"minimumAmount": "30454920"
"router": "<string>",
"includedSwapSources": [
"<string>"
],
},
"destination": {
"inputCurrency": {
"currency": {
"chainId": 8453,
"address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
"symbol": "USDC",
"name": "USD Coin",
"decimals": 6,
"metadata": {
"logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
"verified": false,
```

RELAY

fees object

Learn more about fees

Show child attributes

details object

A summary of the swap and what the user should expect to happen given an input

Show child attributes

protocol object

Protocol information for the quote

Show child attributes


```
            "isNative": false
          }
        },
        "amount": "30754920",
        "amountFormatted": "30.75492",
        "amountUsd": "30.901612",
        "minimumAmount": "30454920"
      },
      "outputCurrency": {
        "currency": {
          "chainId": 8453,
          "address": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
          "symbol": "USDC",
          "name": "USD Coin",
          "decimals": 6,
          "metadata": {
            "logoURI": "https://ethereum-optimism.github.io/data/USDC/logo.png",
            "verified": false,
            "isNative": false
          }
        },
        "amount": "30754920",
        "amountFormatted": "30.75492",
        "amountUsd": "30.901612",
        "minimumAmount": "30454920"
      },
      "router": "<string>",
      "includedSwapSources": [
        "<string>"
      ]
    }
  }
},
"protocol": {
  "v2": {
    "orderId": "<string>",
    "orderData": "<unknown>",
    "paymentDetails": {
      "chainId": "<string>",
      "depository": "<string>",
      "currency": "<string>",
      "amount": "<string>"
    }
  }
```


mintlify